

Project Title

1.EduTrack 360 – Smart Student Progress, QR Portfolio & Certificate Management System

OR

2.Smart Student Progress

Tagline:

"One QR Code, Complete Student Journey."

Problem Statement

In most colleges, faculty members maintain student records using Excel sheets, WhatsApp groups, Google Forms, and paper files.

This creates several problems:

- Assignment records are scattered.
- Microproject submissions are difficult to track.
- Faculty manually calculate completion.
- Students don't know their pending work.
- Activity records are maintained separately.
- Certificate generation is completely manual.
- During accreditation (NBA/NAAC), collecting student achievements takes a lot of time.
- There is no single digital portfolio of the student's complete academic journey.

Proposed Solution

Develop a **Smart Student Progress & Digital Portfolio System** where every student receives a **QR Code ID Card**.

By scanning the QR Code, faculty can instantly access the student's complete profile, including:

- Academic Progress
- Subject-wise Submission
- Assignment Status
- Microproject Status
- Practical Completion
- Attendance
- Cultural Activities
- NSS Activities
- Sports Participation
- Internship Details
- Workshops

- Certifications
- Overall Progress Percentage

Once every required task is completed and approved by faculty, the system automatically generates a **Digital Achievement Certificate**.

Objectives

- Digitalize student submission tracking.
 - Reduce faculty paperwork.
 - Provide real-time submission status.
 - Automatically calculate progress.
 - Encourage student participation.
 - Create a digital student portfolio.
 - Generate certificates automatically.
 - Improve transparency.
-

Complete System Workflow

ADMIN

|

Add Faculty, Subjects, Students

|

Generate QR Code for Student

|

Student Login & Dashboard

|

Upload Assignment / Microproject

|

Status = Pending Approval

|

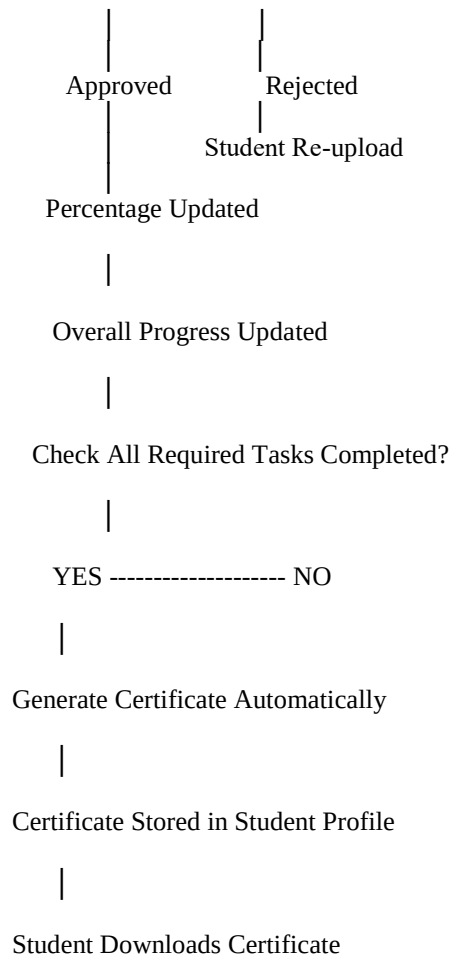
Faculty Login

|

View Pending Submission

|

Approve / Reject Submission



Modules

1. Admin Module

Admin controls the complete system.

Features

- Login
- Add Department
- Add Faculty
- Add Students
- Add Subjects
- Add Assignments
- Add Microprojects
- Add Activities
- Generate QR Codes
- Manage Certificates
- View Reports
- Backup Database

2. Faculty Module

Faculty manages academic activities.

Features

- Login
 - Scan Student QR
 - View Student Profile
 - View Pending Uploads
 - Approve Assignment
 - Reject Assignment
 - Add Remarks
 - Generate Reports
 - View Progress Analytics
-

3. Student Module

Students upload work and monitor progress.

Features

- Login
 - View Subjects
 - Upload Assignment
 - Upload Microproject
 - Upload Activity Certificates
 - View Pending Tasks
 - View Approval Status
 - Track Progress
 - Download Certificate
-

QR Code Workflow

Every student receives a QR Code.

Example:

Student Card

Name

PRN

Department

QR Code

Faculty scans QR.

↓

Student profile opens instantly.

↓

No need to search manually.

Student Dashboard

Welcome

Rahul Patil

Overall Progress



82%

Subjects

Java

80%

DCN

90%

MIC

75%

Attendance

92%

NSS

100%

Cultural

70%

Sports

40%

Extra Activities

80%

Pending

Java Assignment 3

DCN Activity

MIC Report

Academic Module

Each subject contains

Assignments

Example

Java

Assignment	Status
------------	--------

Assignment 1	Approved
--------------	----------

Assignment 2	Approved
--------------	----------

Assignment 3	Pending
--------------	---------

Completion

66%

Microproject

Proposal

Report

PPT

Demo

Viva

Each item receives approval.

Completion shown automatically.

Practical

Faculty marks practical completion.

Example

Practical

8/10

80%

Submission Workflow

Student Side

Login

↓

Choose Subject

↓

Assignment 2

↓

Upload PDF

↓

Submit

↓

Status

Pending

Faculty Side

Pending Upload

↓

Open File

↓

Check Content

↓

Approve

OR

Reject

↓

Add Remark

↓

Save

Student View

Assignment 2

Approved



Progress Updated

Automatic Percentage Calculation

Assignment

Suppose

Total Assignments = 5

Completed = 4

$\frac{4}{5}$

=

80%

Microproject

Suppose

Total Tasks = 6

Completed = 5

$\frac{5}{6}$

=

83%

Subject Completion

Weights

Assignments

40%

Microproject

40%

Practical

20%

Example

Assignments = 80%

Microproject = 100%

Practical = 60%

Final Subject

80×0.40

+

100×0.40

+

60×0.20

=

84%

Overall Academic Percentage

Java

84%

DCN

90%

MIC

78%

Average

84%

Activity Modules

NSS

Activities

- Tree Plantation
- Blood Donation
- Swachh Bharat
- Camp
- Awareness Rally

Faculty approves each activity.

Cultural

Activities

- Dance
 - Singing
 - Drama
 - Anchoring
 - Poster
 - Quiz
 - Rangoli
-

Sports

Activities

- Cricket
 - Chess
 - Kabaddi
 - Football
 - Volleyball
-

Extra Activities

- Internship
- Workshop
- Seminar
- Hackathon
- Patent
- Research Paper
- NPTEL
- Coursera
- SWAYAM
- Industrial Visit

Overall Progress Calculation

Category	Weight
Academic	50%
Attendance	10%
NSS	10%
Cultural	10%
Sports	10%
Extra Activities	10%
Overall Progress	100%

Certificate Generation Workflow

This is the main innovation of the project.

Student uploads work

↓

Faculty approves

↓

Progress Updated

↓

All Subjects Completed?

↓

YES

↓

Attendance Above Required Percentage?

↓

YES



Required Activities Completed?



YES



Overall Progress $\geq$ Required Threshold



Generate Digital Certificate



Store in Database



Student Downloads Certificate



Faculty Can Verify Certificate

Certificate Types

1. Perfect Submission Certificate

Requirement

- 100% Academic Submission
 - All Assignments Approved
 - All Microprojects Approved
-

2. Academic Excellence Certificate

Requirement

Academic Progress $\geq$ 90%

3. NSS Excellence

Highest NSS participation.

4. Cultural Excellence

Highest cultural contribution.

5. Sports Excellence

Outstanding sports participation.

6. Overall Best Performer

Highest overall score in the class.

Student Portfolio

The system becomes a digital portfolio that stores:

- Academic Records
- Projects
- Assignments
- Attendance
- NSS Activities
- Cultural Activities
- Sports Achievements
- Certificates
- Internships
- Workshops
- Extra Courses
- Awards

Students can use this portfolio during placements and higher education applications.

Technology Stack

Frontend

- React.js
- Bootstrap
- HTML
- CSS
- JavaScript

Backend

- Django
- Django REST Framework

Database

- PostgreSQL

Authentication

- JWT Authentication

QR Code

- Python qrcode

File Storage

- Cloudinary / Local Storage

Reports & Certificates

- ReportLab or WeasyPrint

Charts

- Chart.js

Why This Project Is Important

For Students

- Clearly shows pending and completed work.
- Encourages timely submissions.
- Builds a verified digital portfolio.
- Rewards consistent participation with certificates.
- Useful during placements and internships.

For Faculty

- Eliminates manual record-keeping.
- Saves time in tracking submissions.
- One-click access to student progress.
- Easy report generation.
- Faster evaluation and approval workflow.

For College

- Improves transparency and accountability.
- Supports NBA and NAAC documentation.
- Provides centralized academic and activity records.
- Simplifies certificate generation.
- Increases student engagement.
- Enables data-driven academic planning.

Future Enhancements

- AI-based performance prediction for at-risk students.
- Parent portal with progress updates.
- Mobile app (Flutter).
- Email/SMS/WhatsApp reminders for pending tasks.
- Integration with attendance systems.
- Digital badges and gamification.
- Placement readiness score based on achievements.
- QR-verified certificates to prevent forgery.
- HOD and Principal dashboards with department-wide analytics.

Expected Outcome

The final product will be a **complete QR-enabled Student Progress and Digital Portfolio Management System** that centralizes academic submissions, extracurricular achievements, approval workflows, analytics, and automated certificate generation. It reduces manual work for faculty, increases student accountability, and provides every student with a verified digital record of achievements throughout their academic journey.

Additional Smart Features (to make it unique)

-  Green/Yellow/Red progress indicators based on completion.
-  Live dashboard with progress charts for students and faculty.
-  QR code on student ID card opens the profile instantly.
- ☆ Badge system (Assignment Champion, NSS Hero, Cultural Star, etc.).
- ⚡ Semester leaderboard based on overall score.
-  One-click export of student progress to PDF/Excel.
- 🔍 Search students by name, PRN, roll number, or QR scan.
-  Department-wise and class-wise analytics for faculty and HOD.